

Answer Key: Midterm

Sep 18, 2026 ·

Answer key for Doc, based on Doc.

Part 1: Theoretical argument

1. Summarize the theoretical argument.

Men are stereotyped as strong and aggressive, women as weak and passive. Because hurricanes carry gendered names, people expect female-named storms to be less severe than male-named ones. That underestimation leads to them to be less prepared for evacuation etc.. and thus female named storms will cause more deaths.

2. Identify the main hypotheses.

- H1: Female named hurricanes will be more deadly than male named hurricanes.
- H1a: A hurricane with a masculine name will be perceived as more severe than one with a feminine name.
- H1b: female named hurricanes will lead to less preparedness compared to male named hurricanes.

3. Counterargument and a single hypothesis.

- **Era.** Before 1979 all names were female, and older storms were deadlier because of weaker forecasting and warnings. *H: Pre 1979 hurricanes were deadlier than post 1979 hurricanes.*
- **Socioeconomic status.** Deaths may depend on where a storm hits and who lives there. Poverty of the area where hurricane hits may be associated with lower ability to evacuate, people often lack cars, money, or health to leave. *H: The poorer the region where hurricanes hit, the deadlier the hurricane.*

There are many more that could fall into this category: topography, population density, the quality of the warning systems, etc..

Part 2: Measurement of variables

4. Role and level of measurement.

Variable	Role	Level of measurement
MasFem	Main independent	Interval (average of rating items; accept ordinal if justified)
Gender_MF	Main independent, as the binary alternative to MasFem	binary
MinPressure_before	Control (also a moderator in the	Ratio

Variable	Role	Level of measurement
	paper's model)	
Category	None	Ordinal
alldeaths	Dependent	Ratio
NDAM	Control	Ratio

5. A main confounding variable.

- **Year or era.** Names were female-only before 1979, and older storms killed more people because forecasting, warnings, and building codes were weaker. This produces a spurious positive association between femininity and deaths.
- **Location and exposed population.** A storm that hits a densely populated or low-lying area may affect or kill more people.
- **Socio-economic vulnerability.** Poverty, mobile homes, and lack of vehicles raise deaths. The NDAM cannot capture this, since it only measures property value, which is higher in wealthy areas.
- **Rain and flooding associated with the hurricane.**
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6. Sketch the distributions.

alldeaths: any sketch that shows strongly right-skewed distribution . Lots of storms close to zero, then a long thin tail stretching out to 256. The median vastly different from mean.

Part 4: The scatterplot

7. Describe the relationship.

- **Direction and strength:** at most a very weak positive association ($r = .11$) , Most storms have low death tolls whatever the name.
- **Outliers:** Camille (256), Diane (200), Sandy (159), and Agnes (117) are all feminine-named and account for most of the apparent pattern. With two main outliers Katrina and removed
- **Shape of the x-variable:** MasFem is not spread evenly. Names cluster at the masculine end (about 1–3) and the feminine end (about 8–10), with almost nothing in between, so it behaves close to a binary variable. it is a function of the measurement choice... where people land on feminine and masculine names in more binary fashion.
- **Non-linearity and spread:** no clear linear or curved trend. The spread of deaths is wider at the feminine end, partly because there are twice as many feminine-named storms (62 vs. 30).

8. Is the conclusion warranted?

No. It is premature to talk about causation based on a bivariate relationship displayed in the plot. As already identified, there are many alternative explanations that could affect the outcome that are not

factored in. The association between deaths and name of the hurricane in the archival observational study does not address how people actually assessed their vulnerability.

Part 5: Comparing visualizations

9. Type of graph.

- **Visualization A:** a bar chart,
- **Visualization B:** a one-dimensional scatterplot (we call it also a dotplot) per each gender

10. Which is more accurate, which more effective?

Overall, visualization B is both more accurate and more effective.

Feature	Visualization A	Visualization B
Axis	Does not clearly show the abundance of zeros (it does not start at 10)	Shows clearly every storm, including zero deaths
Claim	Title says "twice as many"; actual ratio of means is about 1.7 but medians are equal	More neutral title; with a clear decision to show that median lines show both groups at 5, which is more accurate measure of central tendency.
Order	Alphabetical, order is not helpful.	Grouped by gender, ordered by deaths for clarity
Readability	rotated, unreadable labels, the inclusion of grid adds to noisiness without adding any precision, color scheme is not intuitive,, no legend or axis label, too much chart junk	Only the deadliest storms labeled for clarity, legend provided, using color and shape to add the context for the pre-1979 naming standards. :Log scale is however more difficult to read to non-specialists.

Part 6: The experiments

11. Experiments 1 and 2.

	Experiment 1	Experiment 2
Sample	346 UIUC psychology students, participating for course credit	108 UIUC psychology students, participating for course credit
Dependent variable	Predicted intensity of each hurricane (two items, 1–7:	Perceived riskiness of the hurricane (four items, 1–7)

	Experiment 1	Experiment 2
	intense, strong)	
Treatment	Gender of the hurricane name, varied within subjects: every participant rated 5 male and 5 female names	Gender of the hurricane name, varied between subjects: random assignment to Hurricane Alexander or Hurricane Alexandra
Control(s)	no separate control group,	The unnamed "Hurricane" condition

12. Sampling strategy:

Volunteer.

13. Which problems may occur?

Problem	Check?	Justification
Self-selection	Yes	Participants chose to take part; students who pick this study and very likely of them are coastal residents facing real storms in real time.
Selection on the outcome	No	
Ecological fallacy	I will recognize both no and yes depending on the explanation.	Yes. The archival study has an ecological fallacy when it tries to generalize individual decision-making into an aggregate deadliness of storms. No, within the context of experiments. Explicitly, the goal is to just see how people perceive risks and severity, so that would be no. But you could also make an argument that implicitly they are using the experiments to explain deadliness of experiments and in that case that would be yes.
Hawthorne effect	Yes	Participants knew they were being studied and may have paid unusual attention or guessed the purpose, especially

Problem	Check?	Justification
		in Experiment 1, where they rated both male and female names.
Social desirability bias	Yes	Participants may avoid answers that look sexist, especially in the sample of psychology students who may have been conditioned to think more frequently about gender stereotypes. However, this would likely weaken the observed effect rather than create it.

14. The most important shortcoming.

Self-selection and its effect on the external validity of the entire study: young volunteer students in Illinois, which are answering a hypothetical scenario about storms, are not the people who face real hurricanes in real time, so the finding may simply not reflect residents of hurricane stricken areas, or older adults, or people without the resources to evacuate. The experiment tells us little about actual deaths or real decisions. It may overstate or understate the results compared to real decisions when people must consider many different factors in one.

Further, volunteers who opt into a psychology study may be younger, more educated, and more egalitarian than average, so they may suppress stereotyped judgments. That would **understate** the true effect.

Hawthorne effect: Experiment 1, may make gender differences between names more noticeable and so inflate the effect.